

Frobenius 2.5

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February 18, 2024

Problem 1.

Proof. Let $X = \{x_i | i \in \mathbb{N}_n\}$. $\alpha := Hom_{sets}(X, F) \rightarrow F^{|X|}$ where $f \rightarrow \begin{bmatrix} f(x_1) \\ \vdots \\ f(x_n) \end{bmatrix}$. Let $\beta := F^{|X|} \rightarrow Hom_{sets}(X, F)$ where $(f_1, \dots, f_n) \rightarrow [x_i \rightarrow x_i]$. $\forall (x_1, \dots, x_n) \in F^{|Q|}$, $(\alpha \circ \beta)(x_1, \dots, x_n) = \alpha([x_i \rightarrow f_i]) = (f_1, \dots, f_n)$. $\forall f \in Hom_{sets}(X, F)$, $(\beta \circ \alpha)(f) = \beta((f(X_1), \dots, f(X_n))) = [x_i \rightarrow f(x_i)] = f$. Thus α is a bijection. $\forall f, g \in Hom_{sets}(X, F)$, $\forall a, b \in F$, $\alpha(af + bg) = ((af + bg)(x_1), \dots, (af + bg)(x_n)) = (a(f(x_1), \dots, f(x_n)) + b(g(x_1), \dots, g(x_n)))$. Thus it is vector of $|x|$ elements thus $\dim Hom_{sets}(X, F) = |X|$. Thus $Hom_{sets}(X, F)$ is a vectorspace.

$\forall x \in X \exists \delta_x \in C^X \ni \delta_x(x) = 1$ and $\forall y \in X \ni x \neq y, \delta_x(y) = 0$. Let $a_x \in \mathbb{C}$, suppose $f = \sum_{x \in X} a_x \delta_x = 0$. Then $f(y) = a_y = 0 \implies a_x = 0 \forall x \in X \implies \{\delta\}_{x \in X}$ is linearly independent. There are $\{\delta\}_{x \in X} = |X| \implies \dim span \{\delta\}_{x \in X} = |X| = \dim \mathbb{C}^X \implies \{\delta\}_{x \in X}$ is a basis. $\forall f \in \mathbb{C}^X, \forall y \in X, (\sum_{x \in X} f(x) \delta_x)(y) = \sum_{x \in X} f(x) \delta_x(y) = f(y) \delta_y(y) = f(y) \implies \sum_{x \in X} f(x) \delta_x = f$. $\square$

Problem 2.

Proof. G has the following action on X^C , $g(f) = fg^{-1}$. $\forall g, h \in G, (gh)(f) = f(gh)^{-1} = fh^{-1}g^{-1} = (h \cdot f)g^{-1} = g \cdot (h \cdot f)$. $\forall x \in X, (g \cdot \delta_x)(gx) = \delta_x g^{-1}gx = \delta_x(x) = 1$ and $\forall y \in X \ni y \neq gx, (g \cdot \delta_x)(y) = \delta_x(g^{-1}y) = 0 \implies (g \cdot \delta_x) = \delta_{gx}$.

The representations which map to the permutation matrices in $GL(\mathbb{C}^X)$ for some set X are permutation representations. $\square$

Problem 3.

Proof. Let G act on X , thus the representation is a permutation representation. Thus $\forall g \in G, g \rightarrow$ a permutation matrix. The trace of a matrix is the sum of its diagonal entries. Permutation matrices have 1 and only 1, nonzero which is 1 per row / column. Let A be a permutation matrix, if $A_{i,i}$ is either 1 or 0. If $A_{i,i} = 0$ then it means A doesn't fix e_i and takes $e_i \rightarrow A_i$. If $A_{i,i} = 1$, then e_i is fixed. Thus $tr(A)$ calculates how many basis vectors are fixed by the permutation matrix. Thus tr_X calculates how many basis elements are fixed by a particular element of G . $\square$

Problem 4.

Proof. By the above, $\forall g, h \in G, (g \cdot \delta_h) = \delta_{gh}$. $\forall h \in G, gh = h \implies g = gh h^{-1} = h h^{-1} = e$. Thus $\forall g \in G \ni g \neq e, g$ fixes no elements of G with the action of left multiplication. Thus $\text{tr}(\mathbb{C}^G)(g) = 0$. However e fixes everything, so $\text{tr}(\mathbb{C}^G)(e) = |G|$. Let V be a representation. Thus $\langle \text{tr}_V, \text{tr}_{\mathbb{C}^G} \rangle = \frac{1}{|G|} \sum_{g \in G} \text{tr}_V(g) \overline{\text{tr}_{\mathbb{C}^G}(g)} = \frac{1}{|G|} \sum_{g \in G} \text{tr}_V(g) \overline{\delta_e(g)} = \frac{1}{|G|} \text{tr}_V(1) |G| = \dim V$. Let n be the number of irreducible representations of G .

By previous homework, $C^G = \bigoplus_{i=1}^n V_i^{m_i}$. By previous homework, $m_i = \langle \text{tr}_{V_i}, \text{tr}_{\mathbb{C}^G} \rangle = \dim V_i$, by the above. Thus $C^G = \bigoplus_{i=1}^n V_i^{\dim V_i}$. Thus $|G| = \dim C^G = \sum_{i=1}^n \dim V_i \dim V_i = \sum_{i=1}^n (\dim V_i)^2$. $\square$

Problem 5a.

Proof. $\forall u \in [G, G] \forall g \in G, gug^{-1} = uu^{-1}g^{-1}ug = u(u^{-1}g^{-1}ug) = u[u, g]$ and $[u, g] \in [G, G] \implies u[u, g] \in [G, G] \implies [G, G] \triangleleft G$. $\square$

Problem 5b.

Proof. Let $N \trianglelefteq G \ni G/N$ is abelian. $\forall x, y \in G, xNyN = xyN = yxN = yNxn$ because $gN = Ng$ since $N \trianglelefteq G$ from last year. Thus $xyN = yxN \iff x^{-1}y^{-1}xyN = x^{-1}y^{-1}yxN = N \implies x^{-1}y^{-1}xy = [x, y] \in N$. All the iff are reversible, thus $y^{-1}x^{-1}yx \in [G, G] \implies y^{-1}x^{-1}yx[G, G'] = [G, G'] \implies yx[G, G'] = xy[G, G'] \implies y[G, G']x[G, G'] = x[G, G']y[G, G'] \implies [G, G']$ is abelian.

However the statement also implies $[G, G'] \leq N$, for all normal subgroup N where G/N is abelian. Thus $G \rightarrow G/N$ factors through $G \rightarrow G/[G, G']$ since $[G, G'] \leq N$. By the universal property, $\exists! h : G/[G, G'] \rightarrow G/N$ homomorphism such that G/N is $[G \rightarrow G/[G, G']] \circ h$.

Let $\{N_i\}$ be all the normal subgroups of G such that G/N_i is abelian. Since $[G, G'] \leq N_i$ for all N_i . Then $[G, G'] \leq \bigcap N_i$. Since $[G, G'] = N_i$ for some i , and if there was a $N \leq [G, G']$ then $[G, G'] \leq N \implies [G, G'] = N$. Thus $\bigcap N_i \leq [G, G'] \implies \bigcap N_i = [G, G']$ Thus $[G, G']$ is the smallest normal subgroup for which G/N is abelian. $\square$

Problem 6. Show that the one dimensional reps of G are exactly the same as the one dimensional reps of $G/[G, G']$, and that the number of them is $|G/[G, G']|$.

Proof. Let $\alpha : G \rightarrow GL(V)$ where V is a 1 dimensional representation of G . $GL(V)$ is abelian because it is just F^* . Thus α factors through the map $G \rightarrow G/[G, G']$ by the above. Thus α factors through the representation, $\beta : G/[G, G'] \rightarrow GL(V)$. By a previous homework, all irreducible representations of abelian groups are 1 dimensional. Let k be the number of irreducible reps of $G/[G, G']$. Thus $F^{G/[G, G']} = V_1 \oplus \dots \oplus V_k$ and since $\dim V_i = 1$, by the above $|G/[G, G']| = \dim F^{G/[G, G']} = \sum_{i=1}^k 1 = k \implies k = |G/[G, G']|$. $\square$

Problem 7.

Proof. For a $\mathbb{C}[G]$ module V , it is a module relative to $\mathbb{C}[G]$ and a vectorspace relative to $\mathbb{C}$. G acts linearly on both settings. If V is viewed relative to $\mathbb{C}[G]$ it is a module if it is viewed relative to $\mathbb{C}$ it is a representation.

More formally, let V be a $\mathbb{C}[G]$ module and let $\forall g, h \in G$ and $\forall v, w \in V$ and $\forall c \in \mathbb{C}$. Formally define $\phi(g) : V \rightarrow V$ where $v \mapsto g \cdot v$

$\phi(g)(v + cw) = g \cdot (v + cw) = g \cdot v + g \cdot cw = g \cdot v + cg \cdot w, \phi(g)$ is linear. Because $\phi(1)$ is the identity map and $\phi(g)(\phi(g^{-1})v) = g \cdot g^{-1} \cdot v = v$ thus $\phi(g)$ is invertible. Thus $\phi : G \rightarrow GL(V)$ and thus is a

representation. Similarly given a representation, ϕ , we can do the opposite, take a $f : G \rightarrow \mathbb{C}$ and then do $f \cdot v = \sum_{g \in G} f(g)\phi(g)v = \sum_{g \in G} f(g)gv$, thus we have created a $\mathbb{C}[G]$ module.

$\forall z \in \mathbb{C}(G), \exists f \in \text{Hom}_{\text{sets}}(G, \mathbb{C}) \ni z = \sum_{g \in G} f(g) * g$. Let V be a G -invariant subspace of any G -rep. $z * V = (\sum_{g \in G} f(g) * g) * V = \sum_{g \in G} f(g) * gV \subseteq \sum_{g \in G} f(g) * V$, by G -invariance. Furthermore, by the closure under addition and multiplication properties of subspaces, $z * V \subseteq \sum_{g \in G} f(g) * V \subseteq V$. Thus z preserves the G -invariant subspaces of any G -rep.

Suppose $f(g)$ is a class function, then $\sum_{g \in G} f(g) * g = \sum_{g' \in G} f(hg'h^{-1}) * hg'h^{-1} \forall h \in G$, by properties of class functions and reindexing of G , due to conjugation creating orbits. Thus $\sum_{g \in G} f(g) * g = h \sum_{g' \in G} f(g') * g'h^{-1} \implies zh = (\sum_{g \in G} f(g) * g)h = h \sum_{g' \in G} f(g') * g' = hz$. Thus z commutes with every element of G . $\forall y \in \mathbb{C}[G], y = \sum_{g \in G} c(g)g$ for some $c \in \text{Hom}_{\text{sets}}(G, \mathbb{C})$. $yz = (\sum_{g \in G} c(g)g)(\sum_{g \in G} f(g) * g) = \sum_{g \in G} (c(g)g(\sum_{h \in G} f(h) * h)) = \sum_{g \in G} (c(g)(\sum_{h \in G} f(h) * hg)) = \sum_{g \in G} ((\sum_{h \in G} f(h) * h)c(g)g)$, by the above and the fact that $c(g) \in \mathbb{C}$ for all $g \in G$ and thus commutes. Thus $yz = (\sum_{h \in G} f(h) * h) \sum_{g \in G} c(g)g = zy$. Thus $z \in Z(\mathbb{C}[G])$.

Suppose, $z \in Z(\mathbb{C}[G])$. Then $\forall h \in G, h(\sum_{g \in G} f(g) * g) = (\sum_{g \in G} f(g) * g)h \implies h(\sum_{g \in G} f(g) * g)h^{-1} = \sum_{g \in G} f(g) * g \implies \sum_{g \in G} f(g) * hgh^{-1} = \sum_{g \in G} f(g) * g$. $\forall g \in G$, let $g' = h^{-1}gh$, thus $g = hg'h^{-1}$. Thus $\sum_{g \in G} f(g) * g = \sum_{g' \in G} f(hg'h^{-1}) * hg'h^{-1} = \sum_{g \in G} f(g) * hgh^{-1}$. Thus $\sum_{g \in G} [f(g) - f(hgh^{-1})] * hgh^{-1} = 0$. Since all the elements of G are basis elements of $\mathbb{C}[G]$ they are linearly independent so $\sum_{g \in G} [f(g) - f(hgh^{-1})] * hgh^{-1} = 0 \iff f(g) - f(hgh^{-1}) = 0 \implies f(g) = f(hgh^{-1})$. Thus $f \in \text{Hom}_{\text{sets}}(G/G, \mathbb{C})$ and is a class function. $\square$